Read the Piot engine as an invention

Task 3.1 / Prepare and draw

Name _____ Team _____ Date _____

01 Set up the investigation

Use Fig. 1 of Thomas Piot's 1891 UK patent 20,081 and the teacher demonstration. Identify the heat source, vessel, tubes and surrounding water. Trace the route with arrows.

Open Piot’s original patent and figure (also linked in the lesson)

Trace the engine tubes and mark the waterline on a copy you draw. Pick a fixed travel distance for later tests.

Feature	Purpose inferred	Evidence in figure or demonstration
Vessel		
Tube opening		
Heat source		

02 Plan it on paper

Identify the existing design, the function to preserve, and one measurable improvement. Label each separately.

My prediction, chosen feature or design limit

Read the Piot engine as an invention

Task 3.1 / Test, compare and explain

Name _____ Team _____ Date _____

03 Run the test and record what happened

Observe a teacher-run demonstration. Sketch the motion and identify which explanations are observations and which are inferences. Swap annotated sketches. Have a partner trace water into and out of the mechanism and identify one part needing inspection.

What you saw	What you infer	How to check it
Motion		
Water path		
Possible change		

04 Use the reading

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Complete the assigned checks in the interactive reader. Here, identify a page and one idea that helps explain your result or design decision.

05 Individual explanation

What would count as an improvement on this patent? Give a metric and the starting conditions you will keep.

Keep these sheets with the work. Save the individual explanation in the lesson’s Explain step.